



Operating

Given a customer scenario where changes are made directly in production, explain the implications on the development lifecycle and steps to integrate changes into Application Lifecycle Management.

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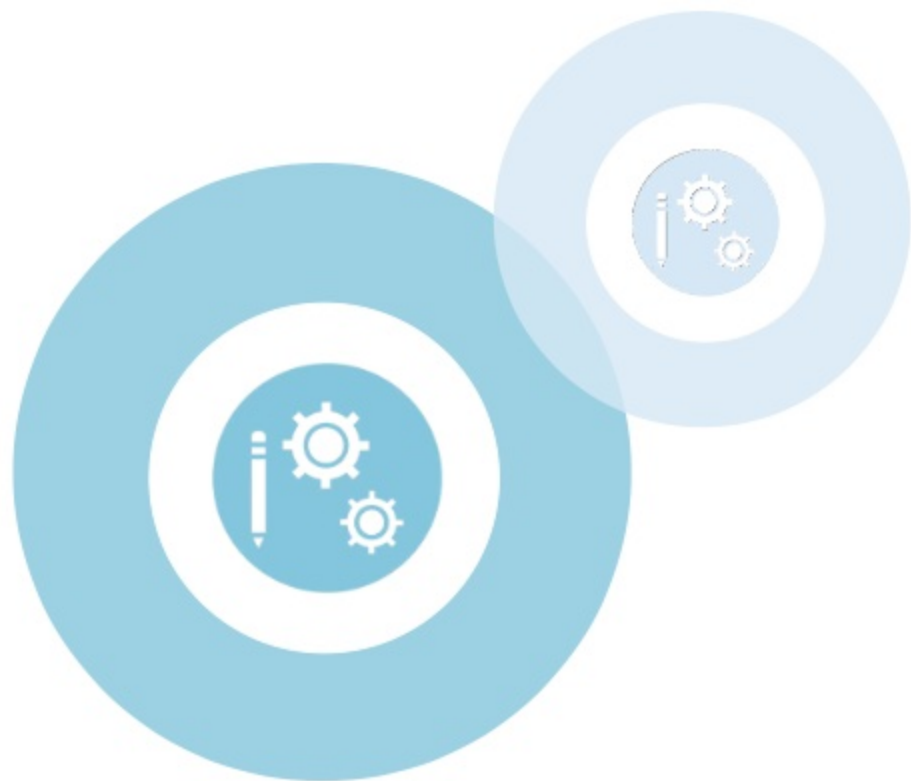
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After studying this topic, you should be able to:

-  Identify the challenges, considerations, and best practices related to direct changes in production.
-  Describe how to integrate direct production changes into Application Lifecycle Management.
-  Recommend appropriate strategies to integrate direct production changes in Application Lifecycle Management.



Introduction

Integrating changes directly into a production environment in Salesforce presents **unique challenges** and **considerations**. It is important to manage these changes within the broader context of **Application Lifecycle Management (ALM)**. Such changes have **implications** on the development lifecycle, from potential disruptions to workflow, system integrity, and compliance issue, making it necessary to utilize **strategies** for effectively incorporating them into the ALM process. The **risks** associated with direct production changes need to be mitigated, ensuring seamless integration, and maintaining the balance between rapid deployment needs and the necessity for rigorous testing and validation processes. Architects should be able to make informed decisions with the aim of upholding system stability and user experience while accommodating urgent business requirements.



Integrating Direct Changes

Risks

Risks associated with direct changes in production include errors, conflicts, and regulatory and compliance issues.



Best Practices

Emergency hotfixes, review, and authorization can be utilized for managing direct changes in production..



Tools

Deployment tools, sandboxes, and sandbox refresh cycles should be used for direct changes in production.



Integration

After direct changes in production, a strategy should be developed to integrate them into the ALM. The changes should be tested as well.



Integrating Direct Changes

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Risks of Direct Changes in Production

Making direct changes in production **without performing testing** has certain risks.



Bypassing established testing protocols can introduce errors into the live environment, **complicating diagnosis and rectification.**



Direct changes risk conflicting with other ongoing updates or integrations, potentially causing **system instability or failure.**



Regulatory and compliance issues may arise if **changes affecting data security or user privacy** haven't been rigorously validated.

Best Practices for Direct Changes in Production

Certain **best practices** can be utilized for **managing direct changes** in production.



EMERGENCY HOTFIXES

A **dedicated process** should be established for **emergency hotfixes**, clearly defining what constitutes an **emergency change** that warrants **direct updates** in production.



REVIEW & AUTHORIZATION

Close collaboration should be fostered between **developers, QA teams, and stakeholders** to ensure changes are thoroughly **reviewed and authorized** before implementation.

Tools for Direct Changes in Production

Release management tools offered by Salesforce can be utilized for **direct changes** in production.



TOOLS

Deployment tools offered by Salesforce and **sandbox refresh cycles** can be leveraged to **plan and execute** changes methodically, **minimizing disruptions** to the development lifecycle. Changes should be **thoroughly documented** and follow a **standard cadence**.



SANDBOXES

Salesforce **sandboxes** should be used for **testing significant changes** to ensure they don't negatively impact live data. **Direct changes** should be reserved for **minor, low-impact updates** after assessing all potential effects.

Integrating Direct Changes into ALM

Direct changes in production **should be integrated** into **Application Lifecycle Management**.



STRATEGY

For direct changes made in production, a **strategy** should be developed to **retroactively incorporate** these changes into the **ALM process**. **Downstream environments** should be **updated** to ensure that they reflect the changes made in production.



VALIDATION

Retrospective analysis and **testing** should be conducted in **sandbox environments** even after direct changes to production, to **validate their impact** and ensure they **align with broader system functionalities**.

Communication & Training

Proper communication and training should be conducted for any **direct changes in production**.



COMMUNICATION

Communication should be **prioritized with the team and stakeholders** about any direct changes made in production, ensuring there is a **clear understanding of the reasons and expected outcomes**.



TRAINING

Training or briefing sessions should be conducted for users affected by the direct changes to ensure **smooth adoption** and **minimize operational disruptions**.

Monitoring & Coordination

Ongoing monitoring, adjustment, and **coordination** are essential for direct changes in production.



MONITORING & ADJUSTMENT

After implementing direct changes in production, the system should be **continuously monitored** for any **unforeseen impacts or issues** that may arise. **Further adjustments** should be made if needed.



COORDINATION

Sandboxes should be used for testing whenever possible, especially for **high-impact changes**. **Direct production changes** should be reserved for situations where **sandbox testing is not feasible** or the **change urgency** justifies it.

Business Scenario

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Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic AI Innovations, an emerging force in artificial intelligence, uses Salesforce to bolster its AI-powered customer service tools and data analysis capabilities. Amidst a surge in demand for its AI solutions, the urgency to push minor updates directly in production arises to swiftly add or modify features. However, the company's development team is aware of the complexities involved in modifying a live environment, especially one that handles vast amounts of sensitive data. The Salesforce architect must advise on the implications of making direct changes in production on the development lifecycle. The architect's guidance will be pivotal in integrating these changes into the company's Application Lifecycle Management strategy, ensuring that updates are methodically planned, tested, and deployed to maintain system reliability and compliance while supporting the rapid evolution of its solutions.

Requirements & Solutions

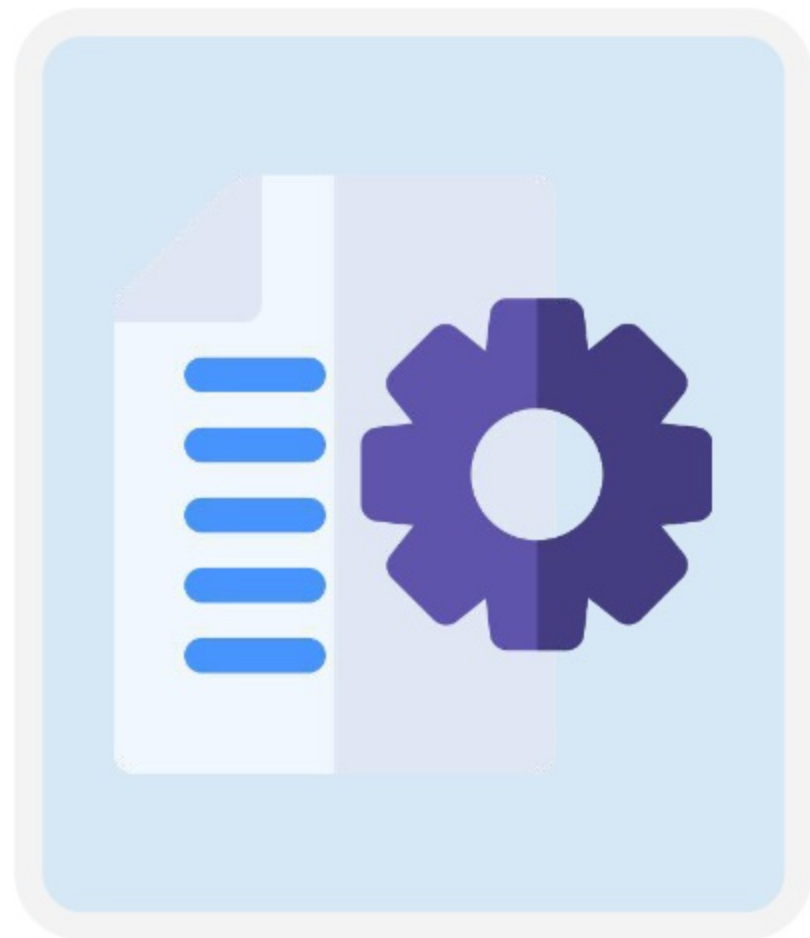
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Requirement 1



SCENARIO

The company is considering the implementation of a minor change policy for low-risk user stories that are commonly handled by a Salesforce administrator. It would allow the administrator to make these changes directly in production. The production environment is highly regulated and audited. The development workflow currently does not utilize continuous integration/continuous delivery (CI/CD). The architect must suggest best practices that the company should follow for the new change policy.

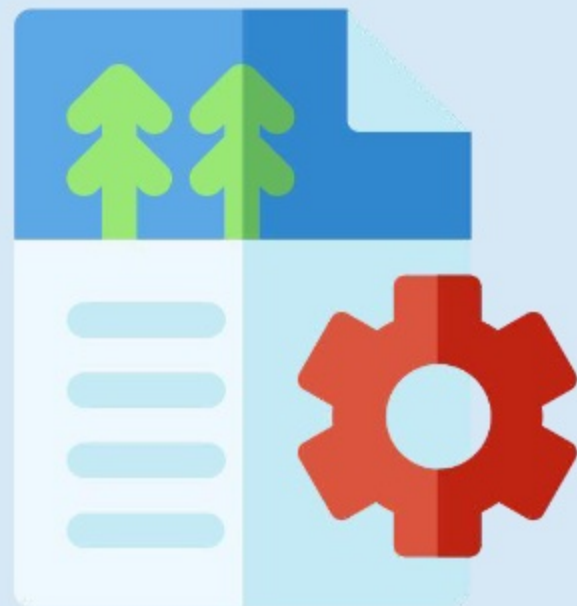


Solution 1



SOLUTION

For the new change policy, minor changes should be **thoroughly documented** and follow a standard cadence. Before making any changes in production, they should be **reviewed** against compliance requirements, regulations, and security principles, and **explicit stakeholder approval** should be obtained. After making the changes, the **downstream environments** should be updated, and the changes should still be **tested** in sandbox environments using the normal development workflow.

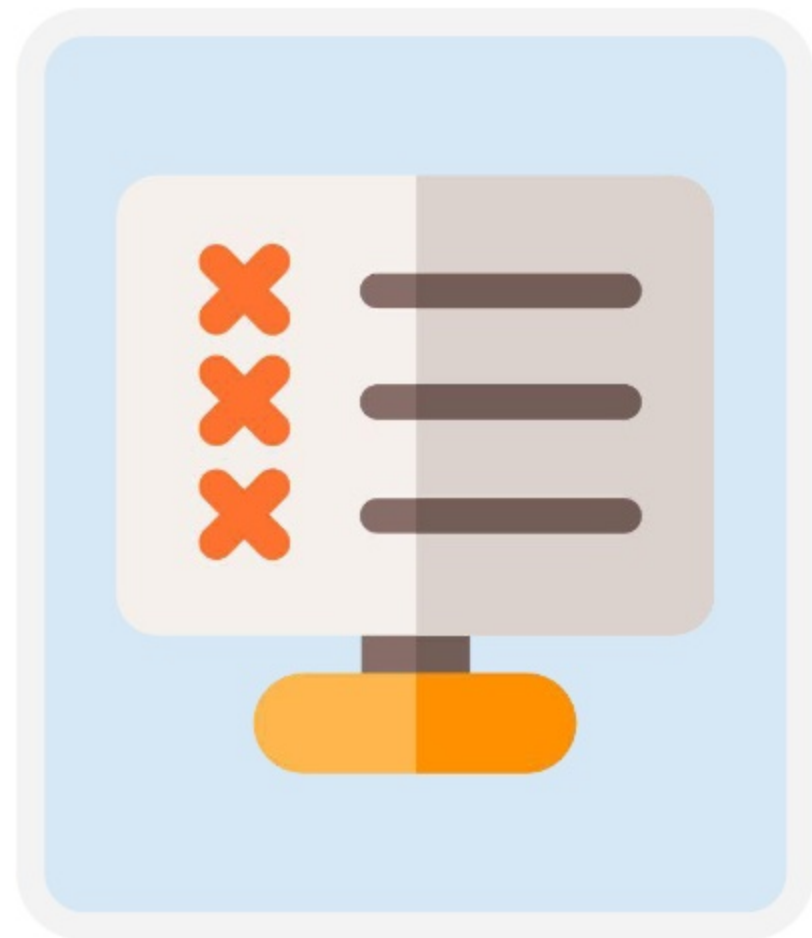


Requirement 2



SCENARIO

A developer has noticed that Apex unit tests are failing in production during deployments, but they still function as expected in the other environments. It has been revealed that a Salesforce consultant has been making minor changes in production while disregarding the need to add dependent components. One of the developers is working in a Developer sandbox on a Lightning web component that utilizes Opportunity fields. The consultant has added a new custom field in production. The architect must provide suggestions to discover the failing unit tests earlier and assist the developer.



Solution 2 (Part 1)



SOLUTION

To help the company discover failing unit tests earlier and manage changes in production more effectively, the consultant should be **trained** to make changes in a **special sandbox**, followed by their deployment to production. Also, a **daily metadata backup** should be committed to a **source control system** and the **differences** should be published so that the development team can **easily spot discrepancies** and **potential causes for test failures** before they affect the production environment.

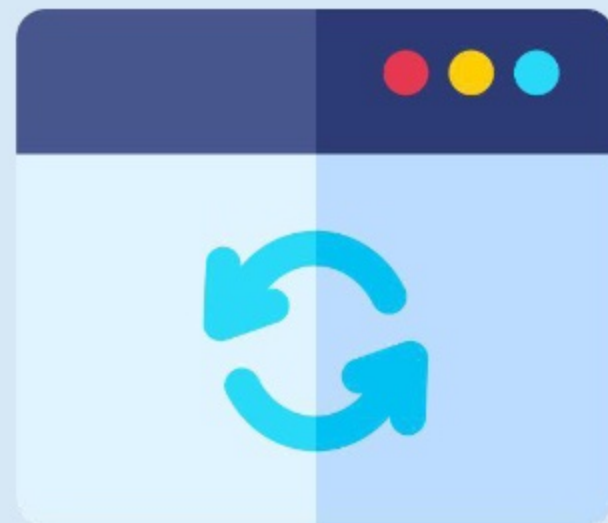


Solution 2 (Part 2)



SOLUTION

To assist the developer in the development process after the addition of a new custom field to production, the consultant should **refresh the Developer sandbox** to ensure that changes made in production are reflected in the sandbox that the developer is using to work on the Lightning web component. Another option is **using a Change Set to deploy the changes from production to the sandbox**.



Learn More

-  [Salesforce Sandbox – Managing Changes in Production](#)
-  [Salesforce Deployment Best Practices](#)
-  [The Difference between Salesforce Production & Sandbox](#)

